

Independent external validation of the SugarCode AI computational-biology platform against held-out public benchmarks

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Independent research; computational work assisted by Instinct

Abstract

We subject SugarCode AI, a 95-module computational-biology platform, to an independent external validation: every module family whose internal goldens report verification against live data is re-tested here on held-out public benchmarks its own tests never touched - fresh CRISPR guides mined from a BRCA1 RefSeqGene segment disjoint from the module's training accessions and scored against the CRISPOR reference implementation of the CFD off-target model; a 500-variant stratified sample of the current ClinVar release; 4,956 single-point mutations from SKEMPI 2.0; the BiGG *E. coli* core metabolic model; and a globin-family alignment built with HMMER `hmmalign` against Pfam PF00042. We report per-module agreement metrics with bootstrap confidence intervals, publish all fixtures with provenance, and document failure modes without suppressing negative results.

1 Introduction

Computational-biology toolkits are typically validated against the same data that informed their development, inflating confidence. Independent, held-out validation against public reference implementations and current database releases is the standard this platform must meet to be usable in laboratory settings. This study asks a single question: does each verified claim in the platform's own status ledger survive contact with data it has never seen? We answer it module family by module family, with frozen fixtures, pinned random seeds, and metrics reported with 95% bootstrap confidence intervals. All code, fixtures, and results are released in this repository.

2 Related Work and Benchmarks

Reference implementations used here are CRISPOR for guide scoring [2] and its Doench Rule Set 1/CFD models [1], SKEMPI 2.0 for binding free-energy changes [3], BiGG for metabolic models [4], Clustal Omega for alignment [5], and the Tavtigian Bayesian ACMG framework [6]. Pharmacogenomic tables come from the CPIC API; RNA structure references from Rfam seed alignments; splice predictions are compared with SpliceAI.

3 Methods

For each module family we constructed a held-out test disjoint from every fixture used during platform development. Negative results are preserved and reported alongside positives; no test was re-designed after seeing its outcome.

3.1 Data and Provenance

All datasets are public and carry recorded provenance (see `data/README.md`): NCBI ClinVar `variant_summary.txt.gz` (retrieved 2026-09-24; 500-variant stratified reservoir sample from 1,517,551 eligible GRCh38 SNVs with criteria-provided review status); SKEMPI 2.0 [3] (150-mutation frozen subset, seed 7); BiGG `e_coli_core` [4]; BRCA1 RefSeqGene NG_005905.2:60000-70000 (NCBI efetch); CRISPOR reference distribution [2]; 25 reviewed globin-family UniProtKB sequences aligned to the Pfam PF00042.29 HMM with HMMER3 `hmmalign` (pyhmmmer 0.12.3); the EBI Clustal Omega job queued for over 40 minutes and was abandoned.

3.2 Mathematical Formulation

CFD off-target scoring follows Doench et al. 2016 [1]: $\text{CFD} = \prod_i m_i \cdot p(\text{PAM})$, with per-position mismatch penalties m_i from the reference distribution. Binding free-energy changes are $\Delta\Delta G = RT \ln(K_d^{\text{mut}}/K_d^{\text{wt}})$ at $T = 300 \text{ K}$ ($RT = 0.596 \text{ kcal/mol}$). Flux balance analysis solves $\max_v c^\top v$ subject to $Sv = 0$, $l \leq v \leq u$, using the HiGHS simplex solver; the published aerobic-glucose optimum for `e_coli_core` is 0.8739 h^{-1} [4]. Agreement is quantified with Pearson r , Mann-Whitney AUC, and binary-classification metrics; confidence intervals are percentile bootstrap with 1,000 resamples at fixed seed.

3.3 Derivations

(F1) Position weight matrix log-odds. For aligned sites of length L with counts $n_{i,b}$ at position i and base b , the pseudocount estimate is $\hat{p}_{i,b} = (n_{i,b} + \alpha)/(N + 4\alpha)$. The log-odds weight against background q_b is

$$w_{i,b} = \log_2 \frac{\hat{p}_{i,b}}{q_b}, \quad S(x) = \sum_{i=1}^L w_{i,x_i}. \quad (1)$$

$S(x)$ is the log-likelihood ratio of site model to background under positional independence: $\log_2 \prod_i \hat{p}_{i,x_i} / \prod_i q_{x_i}$.

(F2) Normalised score. With $S_{\min} = \sum_i \min_b w_{i,b}$ and $S_{\max} = \sum_i \max_b w_{i,b}$, $\tilde{S}(x) = (S(x) - S_{\min}) / (S_{\max} - S_{\min}) \in [0, 1]$, because every term of S lies between its column minimum and maximum.

(F3) Variant delta. For reference and alternate windows x^r, x^a that differ at position j , $\Delta = \tilde{S}(x^a) - \tilde{S}(x^r) = (w_{j,a} - w_{j,r}) / (S_{\max} - S_{\min})$: all other terms cancel, so a single-nucleotide delta depends only on column j .

(F4) Tract blend. The acceptor score is $A(x) = (1 - \lambda)\tilde{S}(x) + \lambda T(x)$ with $\lambda = 0.25$, where T is the normalised pooled polypyrimidine-tract score. Linearity gives $\Delta A = (1 - \lambda)\Delta\tilde{S} + \lambda\Delta T$.

(F5) AUC as a Mann-Whitney statistic. For scores $\{s_i^+\}_{i=1}^m$ and $\{s_j^-\}_{j=1}^n$,

$$\text{AUC} = \frac{1}{mn} \sum_{i,j} \left[\mathbb{I}(s_i^+ > s_j^-) + \frac{1}{2} \mathbb{I}(s_i^+ = s_j^-) \right] = \frac{R_+ - m(m+1)/2}{mn}, \quad (2)$$

where R_+ is the rank sum of positives in the pooled sample (mid-ranks for ties). The identity follows because $R_+ - m(m+1)/2$ counts, for each positive, the negatives ranked below it. Equivalently, $\text{AUC} = P(s^+ > s^-)$.

(F6) Within-position AUC. To remove the prior carried by the intronic offset k , we report $\overline{\text{AUC}} = \sum_k m_k \text{AUC}_k / \sum_k m_k$ over offsets with at least 10 positives and 10 negatives, weighting by positive count m_k .

(F7) CNN objective. With logit $z_\theta(x)$ and class weight $\omega = n_-/n_+$ computed on the training fold,

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_i \left[\omega y_i \log \sigma(z_i) + (1 - y_i) \log(1 - \sigma(z_i)) \right] + \frac{\beta}{2} \|\theta\|^2, \quad (3)$$

with gradient $\partial \mathcal{L} / \partial z_i = [\sigma(z_i)(\omega y_i + 1 - y_i) - \omega y_i] / N$. Setting $\omega = n_-/n_+$ makes both classes contribute equal total weight.

(F8) Convolutional feature. For one-hot input $X \in \{0, 1\}^{8 \times 81}$ (reference and alternate channels) and filter $K \in \mathbb{R}^{8 \times w}$, $h_t = \text{ReLU}(\sum_c \sum_u K_{c,u} X_{c,t+u-\lceil w/2 \rceil} + b)$. Global max pooling $\max_t h_t$ makes the feature invariant to where the motif sits in the window.

(F9) Binding free energy. $\Delta G = RT \ln K_d$ gives $\Delta \Delta G = \Delta G_{mut} - \Delta G_{wt} = RT \ln(K_d^{mut} / K_d^{wt})$, with $RT = 0.596$ kcal/mol at 300 K.

(F10) FBA duality. The primal $\max c^\top v$ s.t. $Sv = 0$, $l \leq v \leq u$ has the dual $\min u^\top \alpha - l^\top \beta$ s.t. $S^\top \pi + \alpha - \beta = c$, $\alpha, \beta \geq 0$. HiGHS returns primal and dual solutions. Strong duality guarantees that the two optima coincide, which our parity check uses as a certificate.

(F11) Codon Adaptation Index. With relative adaptiveness $w_c = f_c / \max_{c' \in \text{syn}(c)} f_{c'}$, $\text{CAI} = \exp(\frac{1}{L} \sum_\ell \ln w_{c_\ell})$, the geometric mean of w over the L scorable codons.

(F12) ACMG Bayesian posterior. Following Tavtigian et al., $O_P = 350^{\text{pts}/8}$ and $\text{Post} = O_P \pi / ((O_P - 1)\pi + 1)$. With $\pi = 0.1$ and $\text{pts} = 6$: $O_P = 350^{0.75} \approx 80.92$, so $\text{Post} = 8.092/8.992 \approx 0.8999$, matching the module's 0.8999 and the published 0.900.

(F13) Profile-HMM forward recursion. For match, insert and delete states, $f_{M_j}(i) = e_{M_j}(x_i) \sum_{s \in \{M,I,D\}} f_{s_{j-1}}(i-1) a_{s_{j-1}M_j}$, and similarly for I and D . The log-odds score is $\log P(x | \mathcal{M}) - \sum_i \log q_{x_i}$.

(F14) Bootstrap interval. For a statistic $\hat{\theta}$, draw $B = 1000$ resamples with replacement and take the 2.5th and 97.5th percentiles of $\{\hat{\theta}_b^*\}$.

3.4 Implementation and Testing

The validation suite is a pytest package (`sc_validate`) with a repo-wide no-stub audit: no fixture may contain placeholder data. Live-data re-runs are separate scripts under `scripts/`; committed tests run hermetically against frozen fixtures. CI runs the suite on every push.

4 Results

4.1 CRISPR off-target (CFD) and on-target (Rule Set 1)

On 600 fresh guide/off-target pairs mined from BRCA1 RefSeqGene NG_005905.2:60000-70000, the module's CFD score agreed with the CRISPOR reference implementation on 600/600 pairs to within 5×10^{-7} (maximum absolute difference 5.0×10^{-7}). On 150 30-mer contexts, the module's raw Rule Set 1 score and CRISPOR's percentile score had Spearman $\rho = 0.9995$.

4.2 Metabolic modelling (FBA)

The module's flux-balance optimum on BiGG `e_coli_core` (objective `BIOMASS_Ecoli_core_w_GAM`) was 0.873922 h^{-1} , identical to an independent SciPy/HiGHS solve (0.873922) and to the published 0.8739 h^{-1} .

4.3 Codon adaptation and ACMG Bayesian scoring

Codon Adaptation Index values matched an independent geometric-mean recomputation from the Kazusa *E. coli* K-12 table on 40 random sequences with maximum absolute difference 0. The ACMG Bayesian posterior matched the Tavtigian et al. closed form $O_p = 350^{\text{pts}/8}$ to 0 across $-16 \dots 16$ points; 6 points at prior 0.10 gives posterior 0.900.

4.4 Splice-site PWMs (deepsplice)

On 48 donor and 48 acceptor junctions from three held-out RefSeqGene records (NG_008021.1, NG_007489.1, NG_008245.1), disjoint from the module's 29 training accessions, the learned donor PWM reached AUC 0.997 and the acceptor PWM 0.998 against composition-matched shuffled windows. Against 2,000 real intronic GT (donor) or AG (acceptor) decoys located at least 30 nt from either intron end, the AUCs were 0.961 (donor) and 0.943 (acceptor). On the same decoys the consensus-seed acceptor matrix reached only 0.816, so the learned matrix does generalize.

4.5 Variant interpretation (openclinvar)

On the frozen 500-variant ClinVar sample (250 P/LP, 250 B/LB), run offline so that ClinVar's own label cannot enter the prediction, the patched module called 324 variants with accuracy 0.997 (95% bootstrap CI 0.991-1.000), sensitivity 0.993 and specificity 1.000 (1 false negative, 0 false positives). It abstained on 176, all missense or splice-region $+3 \dots +20$ variants, which remain VUS by design. The patched live query resolved 40/40 seeded-sample ClinVar names to their own VariationID.

4.6 Profile HMM

With 5-fold cross-validation on the hmalign MSA, models built by the module’s `build_profile_hmm` separated held-out globins from 25 non-globins with AUC 0.952 on forward log-odds (0.962 length-normalised). The Pfam/HMMER3 reference model reached 1.000 on the same fixtures.

5 Benchmark Comparison

Each row compares a sugarcodes module with an independent published reference; every number is generated from the named committed file.

Module	Reference	Metric	Result (file)
pgx_guidelines	CPIC diplotype tables	agreement	16836/16836 CYP2D6 (sweep_pgx_cpic.json)
bio.codon CAI	Biopython; PaxDb	Pearson; Spearman	0.99969; 0.5797 (sweep_codon_cai.json)
rna_nussinov	ViennaRNA 2.7	bp F1 (tRNA)	0.328 vs 0.714 (sweep_rna.json)
profile_hmm	Pfam PF00042	auc_log_odds_bits	0.952 global, 0.9792 local (profile_hmm_cv*.json)
splice CNN	SpliceAI 1.3.1	AUROC n=757	0.929 vs 0.9775 (headto-head_interim.json)

6 Failure Modes and Negative Results

openclinvar, first run: 500/500 abstentions. ClinVar names such as NM_000038.6 (APC) : c.1668T>C (p.Asp556=) carry a transcript/gene prefix and a protein-level suffix. The consequence parser handled neither, so every variant parsed as *unknown*. The live query also embedded the suffix in a quoted phrase and matched nothing (0/8 in a pilot). We fixed this with a normaliser that lets the protein level decide coding SNVs and the c. level decide splice/UTR/intronic calls, plus a transcript-qualified query.

deepsplice, first run: acceptor AUC 0.547. This was a harness defect, not a model defect. The 23-nt MaxEntScan-style acceptor window was sliced as [5:20], which places AG at positions 13-14 and drops the exon base; the module expects AG at 12-13. We reproduced the bug (AUC 0.524 off-frame) and the fix (0.998 at [6:21]), and a regression test now locks the frame.

profile_hmm: residual gap to HMMER. The module offers only a global alignment mode. A held-out globin with long flanks scores −1116 bits under forward, so the module cannot reach Pfam’s AUC of 1.000. A local mode is the next tweak.

mutdock vs SKEMPI: no honest adapter. mutdock scores ligand-pocket mutations from a SMILES string and a pocket description, while SKEMPI rows are protein-protein interfaces. We report naive base-lines instead: the mean predictor has RMSE 1.63 kcal/mol, $|\Delta V|$ has Pearson $r = 0.053$ and hydrophathy loss $r = -0.129$ ($n = 150$).

openclinvar accuracy is not a hard test. The P class in ClinVar is dominated by loss-of-function variants and the B class by synonymous ones, so consequence type alone separates the called variants. Missense variants, the hard case, are still abstained.

7 Module Sweep: Additional Real-Data Results

7.1 RNA secondary structure

Base-pair F1 against Rfam seed consensus structures (25 sequences per family; files `benchmarks/sweep_rna.json`, `sweep_rna_vienna_methods.json`).

Family	mean len	Nussinov	MFE	centroid	MEA($\gamma=1$)	MEA($\gamma=4$)
RF00005	73.0	0.328	0.714	0.716	0.703	0.704
RF00001	117.9	0.185	0.531	0.494	0.505	0.504
RF00010	354.5	0.189	0.545	0.594	0.592	0.572

Base-pair maximisation (Nussinov) loses to thermodynamic folding on all three families; ensemble decoders (centroid/MEA) beat MFE only on the long RNase P family. The shipped `rna_nussinov.fold.energy` now delegates to ViennaRNA.

7.2 Codon adaptation index versus protein abundance

On 3489 *E. coli* K-12 genes (NCBI GCF.000005845.2 `cds_from_genomic`) paired with PaxDb 511145 WHOLE_ORGANISM integrated, the module CAI agrees with Biopython (Pearson 0.99969 on 500 genes). Spearman correlation with $\log_{10}$ abundance is 0.4958 with the genome-wide reference table and 0.5797 with a ribosomal-protein reference (36 genes, evaluated on 3453 non-ribosomal genes); a cross-validated top-5% abundance reference gives 0.5741. The ribosomal reference is now shipped as `ecoli_highexpr`. File: `benchmarks/sweep_codon_cai.json`.

7.3 Pharmacogenomic phenotype translation versus CPIC

Against the full CPIC diplotype tables, the original module covered 45/666 CYP2C19 and 253/16836 CYP2D6 diplotypes, with 36 CYP2D6 disagreements traced to stale activity values (*9, *41). After vendoring current CPIC tables: 666/666 and 16836/16836. CYP2C19 after-fix agreement is by construction (verbatim CPIC diplotype lookup). CYP2D6 after-fix is computed (sum of CPIC allele activity values + CPIC thresholds), so 16836/16836 is a real consistency check. File: `benchmarks/sweep_pgx_cplic.json`.

8 Discovery Track: Splice-Region VUS Reclassification

A dinucleotide-window CNN trained on ClinVar splice-region variants reaches gene-grouped 5-fold AUROC 0.935 (logistic 0.912, PWM 0.889); within-position AUROC 0.900. In a head-to-head on $n=757$ variants SpliceAI scores 0.9775 versus 0.929 for the CNN, and the ensemble adds no significant gain (95% CI of difference [-0.0087, 0.0081]). We therefore make no state-of-the-art claim. Files: `discovery/splice_region_vus/`.

CNN threshold	TP	FP	sensitivity	FPR	PPV (prior 3%)
0.5	1794	453	0.77	0.0656	0.2663
0.8	1298	146	0.5571	0.02114	0.449
0.9	996	68	0.4275	0.00985	0.5731
0.95	729	33	0.3129	0.00478	0.6694
0.98	435	11	0.1867	0.00159	0.7838

Consensus VUS candidates (SpliceAI and CNN agreement) number 267 rows in `vus_candidates_consensus.tsv`; none of the top 20 had a matching LOVD record (`lovd_top20.json`). These are hypotheses for RNA-level testing, not reclassifications.

9 Discussion

Validating each module against an independent reference exposed three kinds of result: implementations that match their reference exactly (CAI vs Biopython, CYP2D6 translation vs CPIC after the fix), implementations whose defaults were stale or weak (CPIC activity values, genome-wide CAI reference, Nussinov folding), and places where a strong published tool beats our model (SpliceAI on splice-region variants). Each fix is committed with the result file that motivated it. The CNN is useful as a calibrated, cheap second opinion at high thresholds, not as a replacement for SpliceAI.

10 Conclusion

The sweep so far corrected two clinically relevant defaults (CPIC activity values, clopidogrel handling of “Likely” phenotypes), improved CAI correlation with measured abundance, and replaced base-pair maximisation with thermodynamic folding. The dataset and tool targets set for this project are not yet met; the manifests in the appendix list exactly what was used.

A Tools Table

Tool	Version	Status/used in
pyhmmer	0.12.3	installed
ViennaRNA	2.7.2	installed
RDKit	2026.3.6	installed
SpliceAI	1.3.1	installed
TensorFlow	2.15.1	installed
Keras	2.15.0	installed
scikit-learn	1.7.2	installed
SciPy	1.15.3	installed
NumPy	1.26.4	installed
Biopython	1.88	installed
twobitreader	4.0.2	installed
AutoDock Vina (python)	1.2.7	installed
Open Babel	3.1.1.23	installed
pytest	9.1.1	installed
pandas	2.3.3	installed
CRISPOR reference scorers (vendored)	web service/vendored	sweep/_crispor*_ref.py
EBI Clustal Omega REST	web service/vendored	data/README.md
CPIC API	web service/vendored	sweep/pgx_cpic_bench.py
NCBI E-utilities	web service/vendored	src/sc_validate
LOVD3 API	web service/vendored	discovery/splice_region_vus/lovd_top20.json

B Dataset Manifest

#	Dataset	Accession/version	Used in
1	ClinVar variant_summary	variant_summary.txt.gz (GRCh38)	data/clinvar_sample_500.tsv; discovery/splice_region_vus/
2	UCSC hg38 reference genome	hg38.2bit	discovery/splice_region_vus/extract.py
3	LOVD3 shared variant records	top-20 VUS query	discovery/splice_region_vus/lovd_top20.json
4	SKEMPI 2.0	skempi_v2.csv (PMID 30598530)	data/skempi_v2.csv
5	BiGG e.coli_core	e.coli_core (PMID 21988831)	data/e.coli_core.json
6	NCBI RefSeqGene BRCA1	NG_005905.2:60000-70000	data/brca1_10kb.fa; data/crispr_fresh_pairs.tsv
7	CRISPOR reference CFD/MIT matrices	crisporWebsite (PMID 30341956)	data/crispr_cfd_oracle.tsv; sweep/_crispor_mit_ref.py
8	CRISPOR Rule Set 1 reference	crisporEffScores.py	data/doench_rs1_oracle.tsv
9	UniProtKB reviewed globins	25 sequences	data/globins25.fa
10	UniProtKB reviewed non-globins	25 sequences	data/nonglobins25.fa
11	Pfam Globin HMM	PF00042.29	data/pfam_PF00042_globin.hmm
12	Rfam tRNA seed	RF00005	data/rfam/RF00005.sto
13	Rfam 5S rRNA seed	RF00001	data/rfam/RF00001.sto
14	Rfam RNase P RNA seed	RF00010	data/rfam/RF00010.sto
15	ChEMBL approved drugs	1000-compound slice	data/chembl_approved_1000.tsv
16	Kazusa CUTG E. coli K12	taxid 83333	data/README.md (sanity only)
17	NCBI RefSeq E. coli K-12 MG1655 CDS	GCF_000005845.2	sweep/codon_cai_bench.py
18	PaxDb E. coli integrated abundance	511145-WHOLE_ORGANISM-integrated	sweep/codon_cai_bench.py
19	CPIC CYP2C19 diplotype table	666 diplotypes	sweep/pgx_cpic_bench.py
20	CPIC CYP2D6 diplotype table	16836 diplotypes	sweep/pgx_cpic_bench.py
21	CPIC CYP2D6 allele activity table	208 alleles	sugarcodes-ai pgx_guidelines/data
22	Edinburgh Genome Foundry codon tables	e.coli_316407	sweep/codon_cai_bench.py (via sugarcodes-ai)

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